

and what model developers can do about them





In a 2020 survey of global organizations, about half of respondents reported using artificial intelligence (AI) in at least one core business function.¹ In healthcare organizations, it's lower. So despite talk of AI becoming a ubiquitous force that's transforming every aspect of our lives, it's telling that about half of the world's businesses aren't even actively using it.

why? | **Several difficult challenges slow organizations' efforts to integrate and use AI in ways that create measurable, practical value for their businesses.**

Many of these challenges cross industries and applications. But in healthcare, they take on distinctive attributes that reflect the requirements, mindsets, and dynamics that are unique to our industry. And of course, many of these factors explain why healthcare lags other industries when it comes to adopting and using AI—despite its immense potential as a clinical tool.²

For data scientists who develop clinical AI models, it's important to understand these headwinds, how they apply specifically to healthcare, and what it will take to overcome them to create successful AI solutions that improve patient outcomes in real-world clinical environments.

How will AI model developers address these five key challenges?

1. Acquiring and normalizing the right data from the right sources
2. Integrating AI models naturally into clinical workflows
3. Scaling AI models to support hundreds of healthcare organizations and thousands of clinicians
4. Building continuous improvement into every AI deployment
5. Earning the trust of frontline clinicians

CHALLENGE #1:

Acquiring and normalizing the right data from the right sources

Data forms the foundation of every successful AI implementation, and data quality correlates more closely with model performance than any other factor. Finding the right sources of clean, accessible, and well-governed data to support AI models is always a challenge—especially in healthcare organizations where data tends to be isolated in disparate, disconnected systems.

AI model developers and data engineers must start by carefully assessing how they can access the data they need to build clinical AI models—and whether enough unique records exist to generate accurate predictions. Then, there's the difficult process of collecting the data, evaluating it, linking it across data sources, and making it available to machine learning tools in ways that drive development of accurate, unbiased predictive algorithms.



Much of this work of collecting, combining, and normalizing data within healthcare organizations falls well outside the scope of where data scientists prefer to focus. Data preparation requires detailed knowledge of how different healthcare organizations collect and store their data—and what protocols and rules govern it. In this scenario, it's essential to assemble a cross-functional team that understands these dynamics and knows how to get the high-quality data from the right sources into the right formats to power successful AI models.

FOR EXAMPLE – to identify pharmaceuticals, some data sources might use the FDA's National Drug Code (NDC) while others might use RxNorm, a different normalized naming system for generic and branded drugs. NDCs often have multiple numbers assigned to the same drug in cases where more than one manufacturer makes the same medication. In this scenario, if a model expects an RxNorm code, but the source system only provides NDCs, the pre-processing engine needs to include a many-to-one crosswalk to reconcile the differences.

Of course, to deploy a well-validated AI model, the same challenges apply. Someone (or some team) needs to know how to obtain the right data from the right sources, and how to

normalize the data so the model will generate a useful prediction—and perform all of this data collection and curation at scale.

CHALLENGE #2:

Integrating your AI models effectively into clinical workflows

Today, there is often a massive gap between mathematical AI models and the people and systems that need to use them. That's true across different industries and disciplines, but it's especially pronounced in healthcare.

To work in a clinical setting, AI solutions need a natural, effective delivery system that's carefully woven into existing healthcare processes. For example, an AI generated risk score can't add value or improve patient outcomes unless it's closely connected to a defined action or established clinical workflow. Without these connections, risk scores and other AI insights can quickly become irrelevant annoyances or distractions—similar to out-of-context drug-drug interaction warnings that pop up so frequently that clinicians simply ignore them.

These connections could take the form of a new care-focused application that makes them easy for clinicians to access and use. They could also consist of thoughtful integrations with existing clinical applications. In either case, the AI delivery system must fulfill the core function of making insights relevant and actionable—and framing predictions in a context that healthcare professionals value and understand.

In other words, model developers need an effective way to translate and insert the output of mathematical algorithms into the language of healthcare.

**CHALLENGE #3:****Achieving consistent AI performance across hundreds of healthcare environments**

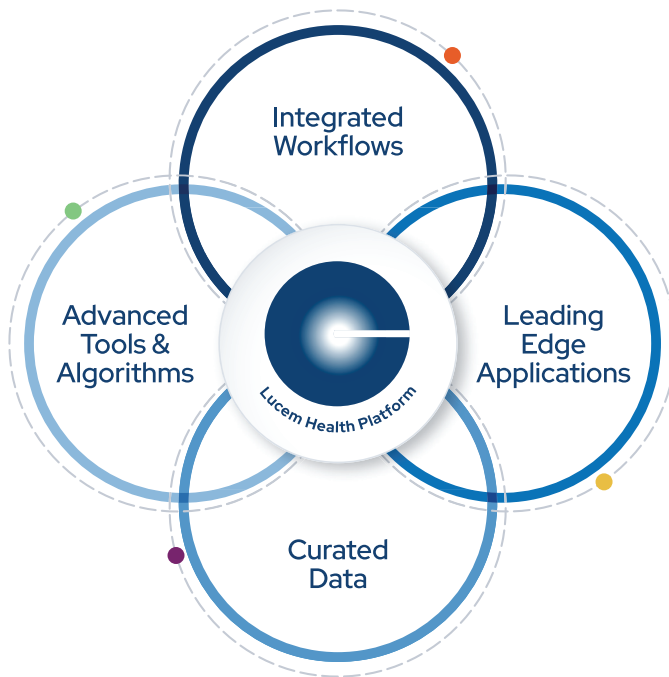
Building a clinical AI model that performs well in one organization using a single dataset is a good start. But to make a positive, meaningful impact on health outcomes (and justify the expense), successful clinical AI models need to perform consistently in hundreds of different environments—each with their own unique mix of IT systems, applications, data, and workflows. So how do you turn a specialized, finely-tuned mathematical model into a practical AI solution that can operate effectively—and generate reliable, accurate insights—for millions of people across dozens or hundreds of different healthcare organizations?

To answer that question, it's important to recognize that data science, software development, and IT deployment are distinct disciplines—with different skillsets, priorities, and areas of focus. To successfully commercialize and scale clinical AI, you need a cross-disciplinary team that can create a standard compute environment for your AI model, connect it to the right data resources, and deploy it to hundreds of healthcare organizations quickly and consistently. If you're a model developer, you could try to assemble and manage that team yourself. But in most cases, it's more efficient to partner with software engineering and deployment experts who already have the technical and industry expertise to deploy, scale, and integrate your model successfully. This approach allows data scientists to focus on what they do best—and avoid deployment distractions and headaches.

CHALLENGE #4:**Building continuous improvement into every AI solution**

As AI models operate in different production environments, they will inevitably encounter unexpected data, unplanned interactions, or new use cases that require adjustments and additional model training. As a clinical AI model becomes more widely deployed and used, it might encounter unfamiliar new patient populations, which can introduce bias and drift—and lead to incomplete or inaccurate results.

To make sure your model remains consistent and accurate after it's deployed, it's important to build an effective monitoring system into the heart of every deployment—one that continuously tests and monitors model performance against a variety of KPIs, detects anomalies, and collects feedback from clinicians—all in real time. This includes creating a continuous feedback loop that makes it easy for data scientists, data engineers, software developers, and IT teams to collaborate on adjustments and improvements. With this kind of ongoing, real-time monitoring system in place, production deployments become more stable and reliable over time.



To unlock the full potential of AI for healthcare organizations, model developers need an open, agnostic platform that can combine data, AI algorithms, applications, and workflows from any system or source.

CHALLENGE #5:

Earning the trust of frontline clinicians and healthcare organizations

It's human nature to mistrust what we don't understand. Highly educated clinicians—trained in the scientific method—are naturally skeptical of any recommendation that's not backed by hard evidence. This creates an interesting "black box" hurdle for AI model developers, because it's difficult or impossible for anyone without advanced mathematical training—including clinicians—to understand exactly how AI algorithms generate insights and recommendations.

One way to overcome this language barrier—and the resulting doubt—involves focusing on the credibility of the outcomes and recommendations clinical AI models produce, rather than trying to explain how they work. For example, clinicians will be more likely to trust an AI model that's validated by proven medical research organizations. That kernel of trust will grow if those endorsements are also backed by credible data that shows AI recommendations are contributing directly to better patient outcomes. And finally, even if clinicians don't have the background to understand exactly how mathematical AI algorithms work, open lines of feedback and communication with model developers—and the sense that their feedback is being taken seriously—can add another layer of trust to the relationship.

Taking these extra steps to validate AI models, track the outcomes and results, and build open lines of communication with clinicians requires additional time and resources, but it's worth the effort. Winning the trust of healthcare organizations and frontline clinicians will ultimately enable AI-powered solutions to create measurable, practical value for provider organizations and their patients.



Maintaining your focus by finding the right partners

These five challenges share one important thing in common:

You don't have to tackle them yourself. With the right approach—and the right partners—you can stay focused on what you do best and then tap into a powerful technology ecosystem and diverse team of experts to fill in the gaps, address key challenges, and deliver the benefits of AI insights to frontline clinicians quickly.

Lucem Health has developed a clinical AI platform that's dedicated exclusively to accelerating AI deployment, adoption, and continuous improvement for model developers and healthcare organizations. This platform is multi-modal and multi-model, which means it can integrate virtually any data with any algorithm—and any algorithm output with any workflow or application. It's neutral, open, and agnostic to meet the diverse needs of model developers across the innovation spectrum. It's built to deploy AI models consistently and reliably to dozens or hundreds of healthcare organizations. It has continuous improvement, clinician and developer collaboration, and trust and adoption baked into its core. It includes tools to help data scientists and frontline clinicians communicate and collaborate.

Finally, it's engineered to deliver the highest possible standards for healthcare data and AI intellectual property protection. When you plug your AI models into this type of platform and supporting ecosystem, you significantly improve the odds that you will deliver meaningful value and transformative outcomes to healthcare organizations around the world.

Lucem Health: The First Agnostic AI Platform for Healthcare

Find out how Lucem Health can accelerate your AI journey—with an agnostic platform that integrates data, AI algorithms, applications, and workflows and delivers practical frontline solutions that transform care delivery and outcomes.

www.lucemhealth.com

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